

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I&II EXAMINATION – SUMMER 2025

Subject Code:BE02000041

Date:19-06-2025

Subject Name:Fundamental of AI

Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Define artificial intelligence and explain its historical origin.	03
	(b) Explain the significance of early expert systems in shaping AI.	04
	(c) Identify real-world applications where early AI principles are still relevant.	07
Q.2	(a) Define an agent and an environment in AI systems.	03
	(b) Compare the scope and capabilities of ANI, AGI, and ASI.	04
	(c) Classify real-world AI applications under ANI, AGI, or ASI categories.	07
	OR	
	(c) Demonstrate how AI has evolved from rule-based systems to machine learning.	07
Q.3	(a) List the different types of knowledge representation techniques	03
	(b) Explain the difference between syntactic and semantic analysis in NLP.	04
	(c) Implement a simple pattern recognition system for recognizing handwritten digits.	07
	OR	
Q.3	(a) Explain the difference between supervised and unsupervised learning with examples.	03
	(b) What are expert systems, and what are their main components?	04
	(c) Apply SVM algorithm for binary classification.	07
Q.4	(a) Define linear regression, logistic regression, and Support Vector Machines (SVM).	03
	(b) Illustrate the difference between training and test datasets.	04
	(c) Provide an example where learning from examples can be used to solve a real-world problem.	07
	OR	
Q.4	(a) Define deep learning and explain its relationship to machine learning.	03
	(b) Explain the differences between convolutional neural networks and recurrent neural networks.	04
	(c) Demonstrate how deep learning is applied to image recognition tasks	07
Q.5	(a) What is a neural network, and what are its main components?	03
	(b) What are Large Language Models (LLMs), and how do they work?	04
	(c) Analyze the role of tools like Bhashini in promoting inclusivity in AI.	07

OR

- Q.5**
- (a)** What are the current issues facing AI development, such as bias, data privacy, and resource consumption? **03**
 - (b)** Compare the capabilities of deep learning with traditional machine learning approaches. **04**
 - (c)** Use ChatGPT to draft a business email. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II EXAMINATION – WINTER 2025****Subject Code:BE02000041****Date:03-01-2026****Subject Name: Fundamental of AI****Time:02:30 PM TO 05:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Define AI. Explain in detail history of AI.	03
	(b) Explain types of AI Techniques.	04
	(c) Enlist & Discuss any two the major task environments of AI.	07
Q.2	(a) What is Pattern Recognition? Give real time example.	03
	(b) Discuss the different approaches to Knowledge representation.	04
	(c) Explain Components & Architecture of an Expert System.	07
	OR	
	(c) What is NLP? Explain working of NLP.	07
Q.3	(a) What is best fit line in linear regression.	03
	(b) Explain Reinforcement learning.	04
	(c) Compare Supervised, Unsupervised & Reinforcement learning.	07
	OR	
Q.3	(a) Compare Classification and Clustering.	03
	(b) Explain Supervised Learning with its types of learning algorithm.	04
	(c) Explain SVM in detail.	07
Q.4	(a) Write note on Deep learning.	03
	(b) Explain Deep Learning Application.	04
	(c) What are the types of Neural Network? Explain any two of it.	07
	OR	
Q.4	(a) Differentiate Deep Learning and Machine learning.	03
	(b) Describe briefly the application of Neural Networks.	04
	(c) Explain Artificial Neural Network in brief with suitable diagram.	07
Q.5	(a) What do you mean by LLMs.?	03
	(b) Write a Short Note on ChatGTP.	04
	(c) Define agent. Enlist the type of agent & explain any two of them.	07
	OR	
Q.5	(a) Briefly explain Rationality with its types.	03
	(b) What are characteristic of Intelligent Agents?	04
	(c) Briefly explain Future Challenges of AI.	07
